🔑 **Answer Key — Teacher Copy · Lesson 5-10**

MSTAR PRACTICE WORKSHEET

Lesson 5-10: Volume of Rectangular Prisms

Grade 6 Mathematics · Standard 6.GR.2

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is the volume of a rectangular prism with l = 4 in, w = 3 in, h = 5 in?

A. 12 in³

B. 24 in³

C. 60 in³

D. 60 in²

V = l × w × h = 4 × 3 × 5 = 60 cubic inches.

Question 2 SELECTED RESPONSE — CORRECT: A

A box has a volume of 36 cm³. Its length is 6 cm and width is 3 cm. What is the height?

A. 2 cm

B. 3 cm

C. 6 cm

D. 12 cm

V = l × w × h → 36 = 6 × 3 × h → 36 = 18h → h = 2 cm.

Question 3 SELECTED RESPONSE — CORRECT: A

How is volume different from area?

- (A) Volume measures 3D space (cubic units); area measures 2D surface (square units)
- (B) Volume uses square units; area uses cubic units
- (C) Volume only applies to cubes; area applies to all shapes
- (D) There is no difference

> Volume measures the space inside a 3D shape (cubic units like in^3). Area measures a flat 2D surface (square units like in^2).

Question 4 SELECTED RESPONSE — CORRECT: A

Two boxes: Box A is 10 × 5 × 4 inches. Box B is 8 × 6 × 5 inches. Which has more volume and by how much?

- (A) Box B by 40 in^3
- (B) Box A by 40 in^3
- (C) Box B by 60 in^3
- (D) They're equal

> Box A: 10 × 5 × 4 = 200 in^3. Box B: 8 × 6 × 5 = 240 in^3. Box B is bigger by 40 in^3.

Question 5 SELECTED RESPONSE — CORRECT: B

What is the volume of a prism with edge lengths 1/2 m, 3 m, and 4 m?

- (A) 3 m^3
- (B) 6 m^3
- (C) 7.5 m^3
- (D) 12 m^3

> $V = 1/2 \times 3 \times 4 = 6\ m^3$.

Question 6 SELECTED RESPONSE — CORRECT: D

A cube has an edge length of 5 cm. What is its volume?

(A) 15 cm³

(B) 25 cm³

(C) 75 cm³

(D) 125 cm³

A cube's volume is edge³ = 5 × 5 × 5 = 125 cm³.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

A container measures 1½ feet long, 2 feet wide, and 3 feet tall. What is its volume?

(A) 6 cubic feet

(B) 6½ cubic feet

(C) 9 cubic feet

(D) 18 cubic feet

1½ = 3/2. V = 3/2 × 2 × 3 = 3 × 3 = 9 cubic feet.

- **A:** The 1½ was rounded down to 1. Use 3/2 for the length: 3/2 times 2 times 3.
- **B:** The three measurements were added. Volume multiplies length, width and height.
- **D:** That is twice the correct volume, as if the length were 3 feet rather than 1½.

Part B — correct answer: C

Why is a mixed number converted to an improper fraction before multiplying edge lengths?

(A) Because improper fractions are always larger.

(B) Because volume cannot be a fraction.

(C) So the whole-number part and the fraction part are both multiplied, not just one of them.

(D) Because mixed numbers cannot be multiplied at all.

Multiplying only the whole-number part drops the fractional part of the edge. Writing 1½ as 3/2 keeps the entire length in the product.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Dropping the Fraction

> Omar found the volume of a box measuring 2½ ft by 4 ft by 2 ft. He wrote: 'V = 2 × 4 × 2 = 16 cubic feet' and said the half 'was too small to matter.'

Explain Omar's misconception and give the correct volume.

Model answer: V = 5/2 × 4 × 2 = 20 cubic feet. The half foot is a real part of the edge; dropping it undercounts the volume by 4 cubic feet.

2 points	Student converts 2½ to 5/2, computes 5/2 × 4 × 2 = 20 cubic feet, and explains that dropping the half discards a real part of the length.
1 point	Student gives 20 cubic feet without explaining the conversion.
0 points	Student agrees with Omar or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A toy company ships action figures in boxes that are 4 × 3 × 6 inches. A shipping crate is 24 × 12 × 18 inches. How many action figure boxes fit in the crate? Explain your reasoning step by step.

Model answer: Each toy box holds 4 × 3 × 6 = 72 in³ and the crate holds 24 × 12 × 18 = 5184 in³, so 5184 ÷ 72 = 72 boxes fit. The division comes out exact because every crate edge is a whole-number multiple of a box edge: 24 ÷ 4 = 6 across, 12 ÷ 3 = 4 deep, and 18 ÷ 6 = 3 layers, and 6 × 4 × 3 = 72 confirms it.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.